RTL

Traccia del 15/02/2021

$IR_x \rightarrow MAR;$ (μ_1)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow T_1;$ (μ_3)
$MAR \rightarrow AC, DECR(T_1) \rightarrow T_1;$ (μ_4)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow A, MAR \rightarrow IND;$ (μ_5)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow B, DECR(T_1) \rightarrow T_1;$ (μ_6)
$A + B \rightarrow T_2;$ (μ_7)
c: **if** $OR(T_1) == 0$ **then**
    $M[MAR] \rightarrow MBR;$ (μ_8)
    $MBR \rightarrow A;$ (μ_9)
    $A + B \rightarrow B;$ (μ_{10})
    $T_2 \rightarrow A;$ (μ_{11})
    $A - B \rightarrow T_3;$ (μ_{12})
    if $T_{3,31} == 1$ **then**
        $B \rightarrow T_2, IND \rightarrow AC;$ (μ_{13})
        $MAR \rightarrow IND, MBR \rightarrow B, DECR(T_1) \rightarrow T_1, INCR(MAR) \rightarrow MAR,$ **goto** c; (μ_{14})
    else
        $MAR \rightarrow IND, MBR \rightarrow B, DECR(T_1) \rightarrow T_1, INCR(MAR) \rightarrow MAR,$ **goto** c; (μ_{15})
    fi
else
    $\varphi;$ (μ_0)
fi

- Check condizione if errata
- Conflitto sul bus dati

$IR_x \rightarrow MAR;$ (μ_1)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow T_1;$ (μ_3)
$MAR \rightarrow AC, DECR(T_1) \rightarrow T_1;$ (μ_4)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow A, MAR \rightarrow IND;$ (μ_5)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow B, DECR(T_1) \rightarrow T_1;$ (μ_6)
$A + B \rightarrow T_2;$ (μ_7)
c: **if** $OR(T_1) == 1$ **then**
    $M[MAR] \rightarrow MBR;$ (μ_8)
    $MBR \rightarrow A;$ (μ_9)
    $A + B \rightarrow B;$ (μ_{10})
    $T_2 \rightarrow A;$ (μ_{11})
    $A - B \rightarrow T_3;$ (μ_{12})
    if $T_{3,31} == 1$ **then**
        $B \rightarrow T_2;$ (μ_{13})
        $IND \rightarrow AC;$ (μ_{14})
        $MAR \rightarrow IND, MBR \rightarrow B, DECR(T_1) \rightarrow T_1, INCR(MAR) \rightarrow MAR,$ **goto** c; (μ_{15})
    else
        $MAR \rightarrow IND, MBR \rightarrow B, DECR(T_1) \rightarrow T_1, INCR(MAR) \rightarrow MAR,$ **goto** c; (μ_{15})
    fi
else
    $\varphi;$ (μ_0)
fi

$IR_x \rightarrow MAR;$ (μ_1)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow T_1;$ (μ_3)
$MAR \rightarrow AC, DECR(T_1) \rightarrow T_1;$ (μ_4)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow A, MAR \rightarrow IND;$ (μ_5)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow B;$ (μ_6)
$DECR(T_1) \rightarrow T_1;$ (μ_7)
$A + B \rightarrow T_2;$ (μ_8)
c: **if** $OR(T_1) == 0$ **then**
    $M[MAR] \rightarrow MBR;$ (μ_9)
    $MBR \rightarrow A;$ (μ_{10})
    $A + B \rightarrow B;$ (μ_{11})
    $T_2 \rightarrow A;$ (μ_{12})
    $A - B \rightarrow T_3;$ (μ_{13})
    if $T_{3,31} == 1$ **then**
        $B \rightarrow T_2;$ (μ_{14})
        $IND \rightarrow AC;$ (μ_{15})
        $MAR \rightarrow IND, MBR \rightarrow B, DECR(T_1) \rightarrow T_1, INCR(MAR) \rightarrow MAR,$ **goto** c; (μ_{16})
    else
        $MAR \rightarrow IND, MBR \rightarrow B, DECR(T_1) \rightarrow T_1, INCR(MAR) \rightarrow MAR,$ **goto** c; (μ_{16})
    fi
else
    $\varphi;$ (μ_0)
fi

- Check condizione if errata

Traccia del 16/07/2020

$IR_x \to MAR, 32 \to T_3$; (μ_1)
$INCR(MAR) \to MAR, 0 \to T_1, 0 \to T_2$; (μ_2)
c: **if** $OR(T_3) == 1$ **then**
$M[MAR] \to MBR, INCR(MAR) \to MAR$; (μ_3)
$MBR \to B$; (μ_4)
if $AC_0 == 1$ **then**
if $B_0 == 0$ **then**
$T_1 \to A$; (μ_5)
$A + B \to T_1, DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_6)
else
$DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_7)
fi
else
if $B_0 == 1$ **then**
$T_2 \to A$; (μ_8)
$A + B \to T_2, DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_9)
else
$DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_{10})
fi
fi
else
$T_1 \to A$; (μ_5)
$T_2 \to B$; (μ_{11})
$A - B \to A$; (μ_{12})
if $A_{31} == 1$ **then**
$T_1 \to AC$; (μ_{13})
else
$T_2 \to AC$; (μ_{14})
fi
fi

$IR_x \to MAR, 32 \to T_3$; (μ_1)
$INCR(MAR) \to MAR, 0 \to T_1, 0 \to T_2$; (μ_2)
c: **if** $OR(T_3) == 1$ **then**
$M[MAR] \to MBR, INCR(MAR) \to MAR$; (μ_3)
$MBR \to B$; (μ_4)
if $AC_0 == 1$ **then**
if $B_0 == 0$ **then**
$T_1 \to A$; (μ_5)
$A + B \to T_1, DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_6)
else
$DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_7)
fi
else
if $B_0 == 1$ **then**
$T_2 \to A$; (μ_8)
$A + B \to T_2, DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_9)
else
$DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_{10})
fi
fi
else
$T_1 \to A, T_2 \to B$; (μ_{11})
$A - B \to A$; (μ_{12})
if $A_{31} == 1$ **then**
$T_1 \to AC$; (μ_{13})
else
$T_2 \to AC$; (μ_{14})
fi
fi

- Conflitto bus dati

$IR_x \to MAR, 32 \to T_3$; (μ_1)
$INCR(MAR) \to MAR$; (μ_2)
$0 \to T_1, 0 \to T_2$; (μ_3)
c: **if** $OR(T_3) == 1$ **then**
$M[MAR] \to MBR, INCR(MAR) \to MAR$; (μ_4)
$MBR \to B$; (μ_5)
if $AC_0 == 1$ **then**
if $B_0 == 0$ **then**
$T_1 \to A$; (μ_6)
$A + B \to T_1, DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_7)
else
$DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_8)
fi
else
if $B_0 == 1$ **then**
$T_2 \to A$; (μ_9)
$A + B \to T_2, DECR(T_3) \to T_3, SHR(AC) \to AC,$ **goto** c; (μ_{10})
else
$DECR(T_3) \to T_3$; (μ_{11})
$SHR(AC) \to AC,$ **goto** c; (μ_{12})
fi
fi
else
$T_1 \to A$; (μ_5)
$T_2 \to B$; (μ_{12})
$A - B \to A$; (μ_{13})
if $A_{31} == 1$ **then**
$T_1 \to AC$; (μ_{14})
else
$T_2 \to AC$; (μ_{15})
fi
fi

- Rottura segnale beta

Traccia del 19/06/2020

$IR_x \rightarrow MAR;$ (μ_1)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow T_1;$ (μ_3)
$DECR(T_1) \rightarrow T_1, 0 \rightarrow T_2;$ (μ_4)
c: **if** $OR(T_1) == 1$ **then**
  $M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
  $MBR \rightarrow B, M[MAR] \rightarrow MBR;$ (μ_5)
  $MBR \rightarrow A;$ (μ_6)
  $A + B \rightarrow B;$ (μ_7)
  $AC \rightarrow A;$ (μ_8)
  $A - B \rightarrow B;$ (μ_9)
  $T_2 \rightarrow A;$ (μ_{10})
  if $B_{31} == 1$ **then**
    $A - B \rightarrow T_2, DECR(T_1) \rightarrow T_1,$ **goto** c; (μ_{11})
  else
    $A + B \rightarrow T_2, DECR(T_1) \rightarrow T_1,$ **goto** c; (μ_{12})
  fi
else
  $T_2 \rightarrow AC;$ (μ_{13})
fi

$IR_x \rightarrow MAR;$ (μ_1)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow T_1;$ (μ_3)
$DECR(T_1) \rightarrow T_1, 0 \rightarrow T_2;$ (μ_4)
c: **if** $OR(T_1) == 1$ **then**
  $M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
  $MBR \rightarrow B, M[MAR] \rightarrow MBR;$ (μ_5)
  $MBR \rightarrow A;$ (μ_6)
  $A + B \rightarrow B;$ (μ_7)
  $AC \rightarrow A;$ (μ_8)
  $A - B \rightarrow B, T_2 \rightarrow A;$ (μ_9)
  if $B_{31} == 1$ **then**
    $A - B \rightarrow T_2, DECR(T_1) \rightarrow T_1,$ **goto** c; (μ_{11})
  else
    $A + B \rightarrow T_2, DECR(T_1) \rightarrow T_1,$ **goto** c; (μ_{12})
  fi
else
  $T_2 \rightarrow AC;$ (μ_{12})
fi

- Conflitto bus dati

$IR_x \rightarrow MAR, 0 \rightarrow T_2;$ (μ_1)
$M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
$MBR \rightarrow T_1, DECR(T_1) \rightarrow T_1;$ (μ_3)
c: **if** $OR(T_1) == 1$ **then**
  $M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$ (μ_2)
  $MBR \rightarrow B, M[MAR] \rightarrow MBR;$ (μ_4)
  $MBR \rightarrow A;$ (μ_5)
  $A + B \rightarrow B;$ (μ_6)
  $AC \rightarrow A;$ (μ_7)
  $A - B \rightarrow B;$ (μ_8)
  $T_2 \rightarrow A;$ (μ_9)
  if $B_{31} == 1$ **then**
    $A - B \rightarrow T_2, DECR(T_1) \rightarrow T_1,$ **goto** c; (μ_{10})
  else
    $A + B \rightarrow T_2, DECR(T_1) \rightarrow T_1,$ **goto** c; (μ_{11})
  fi
else
  $T_2 \rightarrow AC;$ (μ_{12})
fi

- Decremento del valore non aggiornato di T1